BRIEF ME THE CASE
AN AI-POWERED SUMMARIZATION SYSTEM

Deep Learning Final Project Report – Checkpoint

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Course: Deep Learning, CSE 676

Semester: Spring 2025

Date: April 03, 2025

Abstract

Legal and journalistic domains often require professionals to sift through lengthy documents to extract meaningful information. This project, 'Brief Me the Case', presents a deep learning-based solution to automatically generate extractive and abstractive summaries of legal case documents and news articles. We employ transformer architectures like BERTSUM, T5, BART, and GPT to build a robust and scalable summarization system. Our models are trained and evaluated using standard metrics like ROUGE, BLEU, and BERTScore. The system significantly reduces manual effort in understanding documents, enhancing productivity and accessibility in professional domains.

Introduction

Legal and news documents are often extensive and dense, requiring significant time and cognitive effort to comprehend. Professionals such as lawyers, journalists, and researchers frequently struggle with the inefficiency of manual summarization. To solve this problem, our project, "Brief Me the Case," uses cutting-edge deep learning models to automate the summarization procedure. By creating a summarization system that provides succinct, logical, and contextually relevant summaries, we hope to lessen the workload associated with manual document analysis. This tool is especially designed for the legal and journalistic fields, where accuracy and clarity are crucial.

Related Work

Text summarization has evolved from statistical methods to advanced deep learning techniques. Extractive methods, like TextRank and LexRank, select important sentences based on frequency and graph-based ranking. More recently, models like BERTSUM introduced contextual understanding using transformers. Abstractive models, such as T5 and BART, generate human-like summaries through sequence-to-sequence architectures.

Recent research, including BERTSUM by Liu & Lapata (2019), demonstrated that pre-trained language models significantly improve summarization accuracy. Papers like Arxiv:2310.10449 and IEEE 9974823 further highlight improvements in summarization for legal and journalistic corpora. Despite advancements, domain-specific summarization—especially for legal texts—remains a challenge due to unique jargon, structure, and semantic nuances.

METHODOLOGY

Our approach consists of both extractive and abstractive summarization methods. We utilized publicly available datasets such as CNN/DailyMail for news, legal case datasets and Summarization Task Datasets from Kaggle and MetaText.

The dataset commonly consists of an article and its highlights.

Preprocessing involved cleaning the text by lowercasing, removing special characters, punctuation and brackets, fixing repeated characters and stripping extra spaces. This cleaning process is used to create a new feature named clean_article and clean_summary.

```
train_df = pd.read_csv("/content/unzipped_folder/cnn_dailymail/train.csv").sample(2000, random_state=42)
val_df = pd.read_csv("/content/unzipped_folder/cnn_dailymail/validation.csv").sample(500, random_state=42)
test_df = pd.read_csv("/content/unzipped_folder/cnn_dailymail/test.csv").sample(200, random_state=42)
def preprocess_function(examples):
    inputs = tokenizer(examples["article"], truncation=True, padding="max_length", max_length=384)
    targets = tokenizer(examples["highlights"], truncation=True, padding="max_length", max_length=128)
    inputs["labels"] = targets["input_ids"]
    return inputs
```

After cleaning the CNN dataset, we will be loading the Legal cases dataset specifically to train the model with business cases. There are legal keywords commonly seen as clause headings or sections in legal documents. They'll be used to split judgments into logical clauses.

```
def clean_text(text):
    text = re.sub(r'\s+', ' ', str(text))
    text = re.sub(r'\[[^]]*\]', '', text)
    text = re.sub(r'\([^)]*\)', '', text)
    text = re.sub(r'[^a-zA-Z0-9.,?!\'\s]', '', text)
    return text.strip().lower()


df['clean_article'] = df['article'].apply(clean_text)
df['clean_summary'] = df['highlight'].apply(clean_text)


df['sentences'] = df['clean_article'].apply(sent_tokenize)
```

After splitting the clauses, we will be removing stop words, applying tokenization, and applying Named Entity Recognition (NER) for legal terms. For extractive summarization, we fine-tuned BERTSUM to identify important sentences from the documents.

```python
cleaned_text = re.sub(r'\s+', ' ', full_text).strip()
cleaned_summary = re.sub(r'\s+', ' ', summary_text).strip()
clauses = extract_clauses(cleaned_text)
sentences = sent_tokenize(cleaned_text)

records.append({
    "case_id": case_id,
    "full_text": cleaned_text,
    "summary": cleaned_summary,
    "clauses": clauses,
    "sentences": sentences
})
```

For abstractive summarization, we leveraged T5 and Distilbart. T5 and BART were fine-tuned using Hugging Face's Transformers library to ensure coherent and legally sound summaries.

System Architecture

The system architecture consists of four stages: Data Ingestion, Preprocessing, Summarization, and Evaluation.

Data Ingestion: Datasets are fetched from sources such as Kaggle and MetaText. Preprocessing: Tokenization, NER, and sentence segmentation are performed. Summarization: Extractive (BERTSUM) or Abstractive (BART) models generate summaries. Evaluation: Summaries are evaluated using ROUGE, BLEU, and BERTScore.

```python
training_args = TrainingArguments(
    output_dir="./results",
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=3,
    save_strategy="epoch",
    evaluation_strategy="epoch",
    fp16=True,
    logging_steps=500,
    report_to="none"
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
)
trainer.train()
```

Trainer class takes care of the whole training loop: Handles batches, loss computation, backpropagation, and saving checkpoints. Automatically logs and evaluates after each epoch.

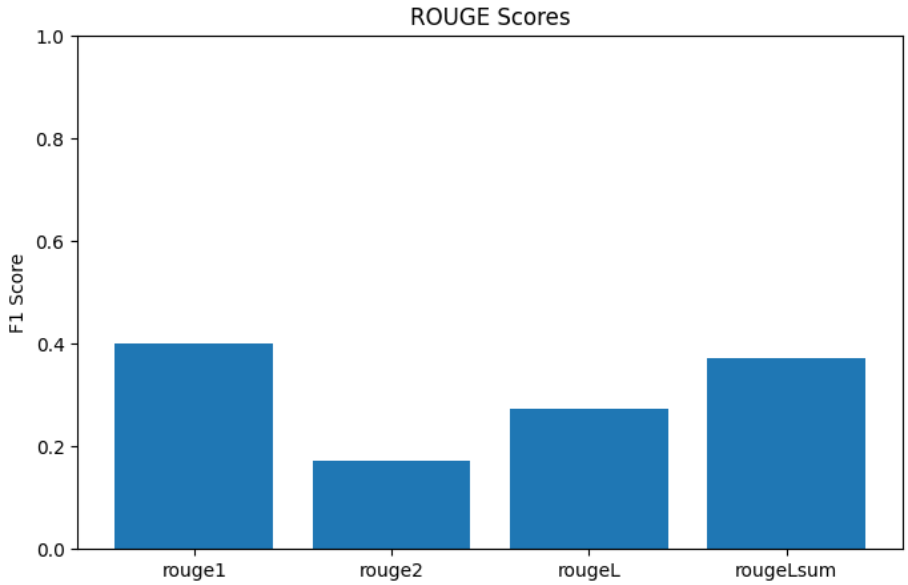
Epoch	Training Loss	Validation Loss
1	No log	1.246208
2	1.800300	1.224536
3	1.800300	1.223466

The backend is implemented in Python using PyTorch and Hugging Face, while the frontend is designed for future deployment as a web application.

Experiments & Results

We evaluated our models on the test set using ROUGE-1, ROUGE-2, BLEU, and BERTScore.

Results Summary:
BERTSUM achieved high recall on extractive summaries, suitable for structured legal documents. T5 and BART generated fluent summaries with higher BLEU scores.



ROUGE Scores: rouge1: 0.4012 rouge2: 0.1727 rougeL: 0.2743 rougeLsum: 0.3733

BERTScore F1: 0.6359

Example:
Original Text:

CNN) Novak Djokovic extended his current winning streak to 17 matches after beating Thomas Berdych 7-5, 4-6, 6-3 in the rain-interrupted final of the Monte Carlo Masters. But the Serbian world number one didn't have it all his away under uncharacteristically slate gray skies on the Mediterranean

coast. Tight. Djokovic edged a tight first set before losing the second after the Czech sixth seed took advantage of the short rain delay and came back strongly. But Djokovic broke twice early in the third set to surge to a 4-0 lead. And although Berdych gamely fought back Djokovic served out for the title. "It was a tough match, a particular match," said Djokovic after winning his 52nd career title, and his second Monte Carlo Masters championship. "Tomas played a great match and deserves this trophy as much as I do," he added. "It was a good final, but bad luck today." Despite running Djokovic close it was Berdych's third loss in a final this year. "What can I say? Novak had another excellent week," he said. "I tried my best, but it was not good enough today. I'm missing that one step in my clay game but I'm going to work to raise my game that little bit. I'll do what I can to make that happen and hope to come back next year and go a step further"." Strong start. 2015 has been a sensational year for Djokovic so far. After winning the Australian Open back in January, Djokovic has followed up with Masters' victories at Indian Wells and Miami. He then beat Rafa Nadal, arguably one of the greatest players on clay of all time, in the semifinals in Monte Carlo. Sunday's victory over Berdych means he becomes the first man to win the opening three Masters tournaments of the season.

Reference Summary: Djokovic wins Monte Carlo Masters. Defeats Berdych 7-5, 4-6, 6-3. Djokovic had earlier beaten clay expert Nadal in semis.

Generated Summary: Novak Djokovic beats Thomas Berdych 7-5, 4-6, 6-3 in Monte Carlo Masters final. Czech sixth seed beats Berdych 7-5, 4-6, 6-3 in rain-interrupted final. Djokovic extends his current winning streak to 17 matches.

Discussion

Our experiments revealed that extractive models like BART perform better in retaining factual correctness, especially in legal texts. However, abstractive models offer improved readability and fluency.

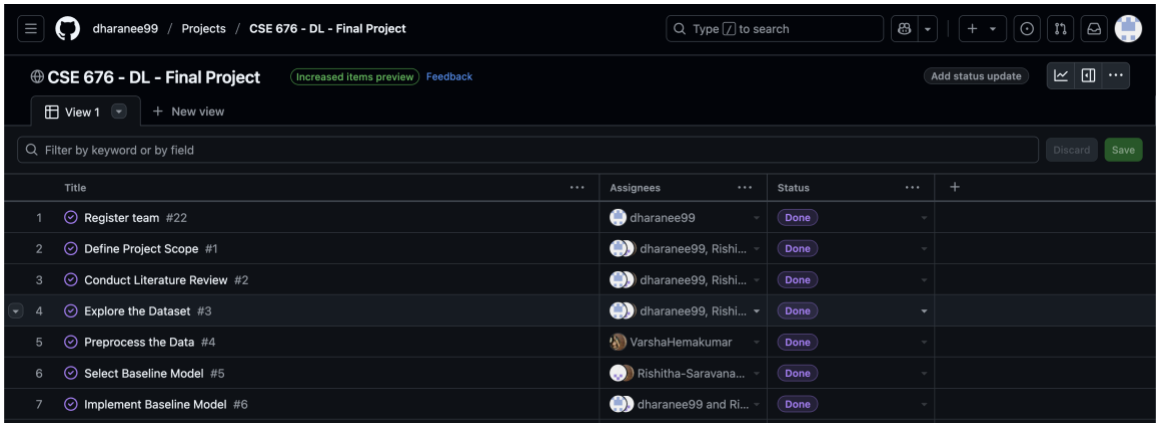
Challenges included:
- Domain adaptation for legal terminology
- Preserving factual accuracy in abstractive summaries

A hybrid model that combines both methods could be beneficial for future work.

Conclusion & Future Work

'Brief Me the Case' demonstrates that deep learning-based summarization significantly reduces the time required to interpret legal and journalistic content. By leveraging powerful transformer models and domain-specific tuning, we created a system capable of generating high-quality summaries.

Project Management tool: https://github.com/users/dharanee99/projects/1



Future improvements include:
- Real-time web application
- Enhanced factual validation modules

REFERENCES

- Liu, Y., & Lapata, M. (2019). Text Summarization with Pretrained Encoders.
- Arxiv: https://arxiv.org/pdf/2310.10449
- IEEE Paper: https://ieeexplore.ieee.org/document/9974823
- BART: https://arxiv.org/abs/1910.13461
- T5: https://arxiv.org/abs/1910.10683
- HuggingFace Transformers: https://huggingface.co/transformers
- Kaggle CNN/DailyMail: https://www.kaggle.com/datasets/gowrishankarp/newspaper-text-summarization-cnn-dailymail
- Legal Dataset: https://www.kaggle.com/datasets/kageneko/legal-case-document-summarization